Lesson 1: figures and sources

Reading Environment and History through Tree Rings / Introductory course / English v5. Slide source labels link to the original papers, image descriptions and method guidance. This list supports review after the first class.

Slide 6: real wood specimens

Buried wood, excavated wood and historic timber, reused from the lecturer's supplied 2021 presentation, slide 30. Images retain their original aspect ratios. The supplied file is recorded as the source without assigning additional photographer identities.

Dendrochronology in archaeology: National Park Service

Slide 11: charred maize from Ceibal

The photograph comes from the supplied 2021 presentation, slide 92. It shows a real specimen, separate from fictional exercise samples A-D. No exercise date is assigned to the photographed material.

Related dating research: Inomata et al. (2017)

Radiocarbon samples and interpretation: Oxford

Radiocarbon calibration: Oxford

Slide 14: measured Yanase sugi series

Yonenobu, H., Hoshino, Y. and Ohyama, M. (2021). Yanase sugi (Cryptomeria japonica) tree-ring chronology 1768-2004. Japanese Journal of Historical Botany 29(1), 15-19. The original figure was reused from the supplied earlier lecture, slide 34, with axes and scales preserved.

Grey lines show individual series and the thick black line the combined chronology. The lower panel shows agreement in year-to-year changes. The ring-width axis is logarithmic. These are measurements from 18 discs, not simulated teaching data.

Original article and PDF

Learning crossdating: University of Arizona

Natural archives and further reading

Slide 15, left: Ichinomegata sediment

The photograph comes from the supplied 2021 presentation, slide 69. A non-destructive crop shows the sediment layers while retaining the original photograph in the presentation. The photograph alone does not establish annual formation.

Slide 15, right: GISP2 ice core

NOAA / USACE photograph of a 19 cm section of GISP2 ice core from a depth of 1855 m. The Commons description identifies it as a US federal government work in the public domain. Image content and aspect ratio are preserved.

Image description and rights

Ice-core overview: Wikipedia

Slide 27: wood cross-section

Tree rings.jpg by Arnoldius, CC BY-SA 2.5. The image content is unchanged and its original aspect ratio is preserved. This is a separate specimen from the Yanase chronology. Individual image licences remain applicable independently of the teaching material as a whole.

Image and author

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Concepts and open data

Dendrochronology: Wikipedia

Chronological dating: Wikipedia

Scientific dating methods: Historic England

Paleoclimate archives: NOAA

International Tree-Ring Data Bank

Figures on slides 4, 5, 8, 10 and 17 are illustrative scenarios or ideal models. The specimen photographs and measured chronology have separate empirical sources.